

Effectiveness of waste prevention program in primary students' schools

Antonis A. Zorpas¹ · Irene Voukkali² · Pantelitsa Loizia²

Received: 19 November 2016 / Accepted: 30 March 2017 / Published online: 20 April 2017
© Springer-Verlag Berlin Heidelberg 2017

Abstract Even though reducing waste is at the top of the waste hierarchy, no real decoupling between waste generation and consumption has been demonstrated. Several waste directives had been published from EU, but they have only brought minor changes within the key objective of reducing waste generation. Most efforts have been targeted towards greater amounts of recycling and better management of waste disposal. While these are necessary and socially beneficial goals, they are not adequate for the achievement of long-term sustainability goals. The purpose of this study is to understand students' knowledge, attitudes and behavioural changes in relation to the water plastic bottle of 500 ml. Understanding waste prevention behaviour (WPB) could enable schools' principals, local authorities and committees as well as decision makers to design and implement more effective policies for reducing the amount of specific waste streams that is generated. Students in a daily base bring their own water containers of 500 ml or buy water from the school as they do not feel safe to use other sources of water. Nine hundred ninety-eight refilling

stainless steel water refilling bottles (SSWRB—of 600 ml) were shared to the students in four primary schools. The results indicated that the students are presented with different behaviours from class to class for many reasons; most of them are related with what their parents believe, and how themselves or the synergies between them reacts and affected.

Keywords Social behaviour · Student's attitude · Monitoring and evaluation · Elementary schools · Environmental education · Prevention activities · Waste prevention · Circular economy

Introduction

Education and knowledge is consider to be one of the most powerful, well-known and proven drivers for sustainable development. The United Nations Conference on Environment and Development (UNCED), held in Rio de Janeiro during 1992, enrolled Agenda 21, which emphasized the potential role played by education in generating the awareness essential to protect the environment and contribute to sustainable development (Rickinson et al. 2009; Kanyimba et al. 2014; Voukkali et al. 2014). As described in chapter 36 of the Agenda 21 “education is critical for promoting sustainable development and achieving environmental and ethical awareness, values and attitudes, skills and behaviour consistent with sustainable development and for effective public participation in decision making” (Voukkali et al. 2014). The World Summit on Sustainable Development (WSSD), held in Johannesburg in 2002, suggested to implement Resolution 57/254, which confirmed the 2005–2014 period to be the United Nations Decade of Education for Sustainable Development (UNDESD) (Haigh 2009). The main vision of the UNDESD was to develop a world where all humans will had the chance to benefit from

Responsible editor: Philippe Garrigues

✉ Antonis A. Zorpas
antonis.zorpas@ouc.ac.cy; antoniszorpas@yahoo.com

Irene Voukkali
irenevoukkali@envitech.org; voukkei@yahoo.gr; <http://www.envitech.org>

¹ Cyprus Open University, Faculty of Pure and Applied Sciences, Environmental Conservation and Management, Lab of Chemical Engineering and Engineering Sustainability, P.O.Box 12794, 2252, Latsia Nicosia, Cyprus

² Department of Research and Development, I.E.S.T–Envitech Ltd (Institute of Environmental Technology and Sustainable Development, P.O.Box 34073, 5309 Paralimni, Cyprus

education as well as to learn about the principles focuses on environmental protection. The UNDESD expressed a commitment to integrate Education for Sustainable Development (ESD) at all levels of training systems and education (UNESCO 2006). The main objective of the eco-education and eco-learning target programs is not only to increase conscious eco-citizens who realize and comprehend the nature as a whole, develop a critical approach in their interaction and collaboration with environment, and are sensitive about environmental topics, aspects and issues, but also to help individuals figure out, analyse, and make researches about environmental issues as well as in their efforts to solve the problems and take decisions as to the matter. Moreover, according to the Waste Framework Directive (WFD) (European Parliament and the Council of the European Union 2008; Lasaridi et al. 2014), “waste prevention are the measures taken before a substance, material or product has become waste, that reduce the quantity of waste, the adverse impacts of the generated waste on environmental and human health or the content of harmful substances”. The government in the UK has sponsored a large research program focus on waste prevention. It includes a review of evidence analysing the behavioural occasions and barriers in household waste prevention, related with the effectiveness and efficiency of various policy actions and measures (Read et al. 2009; Cox et al. 2010). The influence and impact of waste prevention awareness campaigns, and methods to measure, evaluate and monitor waste prevention through mass reduction and behavioural studies (Sharp et al. 2010; Zorpas and Lasaridi 2013; Zorpas et al. 2014; Zorpas et al. 2015) were also developed and examined in details. Existing decoupling indicators of waste prevention were also reviewed (Fell et al. 2010). Decoupling indicators cover the process of separating economic growth from related adverse environmental impacts. It is, however, quite critical also to assess the environmental significance of waste prevention as this can reinforce evidence for policy development. Moreover, according to the same research (Fell et al. 2010) waste prevention should not be a typical general program indicated several prevention activities which may confused the citizens but should focus on individual waste streams and/or behaviours each of which will need a bespoke approach while decoupling should “borne in mind” as backdrop. Moreover, according to Testa et al. (2014a), the production system which provides products and services to the market and consumers is the main factor of affection sustainability.

The model of ESD has also been established in educational policies around the world to rearrange environmental educators contributions to sustainable development (Jickling and Wals 2008; Foo 2013; Hume and Barry 2015). Nevertheless, the adoption of a sustainable development view in education offerings major shortcomings involving to anthropocentric bias and a rejection of the intrinsic value of nature (Selby 2006; Kopnina 2012a, b). Such shortcomings to ESD can be offset by applying an EMS (Environmental Management System) in

education. According to Testa et al. (2014b) and Daddi et al. (2015) an EMS is consider to be a worldwide tool which can be applied by any organization aiming to improved and reach a continuous improvement of environmental performance. This will support a learning process implemented by a school organisational structure regarding social, environmental and economic factors. Furthermore, humans have started to understand different environmental problems including global warming, soil and water, pollution of air, acid rains and an awareness as to the urgency of protecting the environment has arisen (Nag and Vizayakumar 2005). Problems and issues related to environment as mentioned above, not only turn the world into an uninhabitable place, but also intimidate human existence. Could be possible through environmental education (de Kumar 2004) to help citizens to realize the seriousness of the situation without promoting any panic. The main target and objective of the environmental education is not only to raise aware eco-citizens who comprehend the nature as a whole, to be sensitive about environmental topics and improve a critical approach in their interaction with environment, but mainly to assist individuals figure out and do researches about environmental issues (Godwell and Ekpe 2011). In this regard, the significance of environmental education to prevent and solve environmental problems and to raise sensitive and conscious citizens about environment comes into prominence (Foo 2013; Kopnina 2014). Additionally, schools have a great role to play providing this education. The personal approached and attitudes of teachers at schools touch and affect the responses of the students too. Hence, it is essential for teachers to be an exemplary model. The behaviours of teachers as a decent model affect the knowledge, attitudes, awareness and behaviours of students about environment in a positive way (Littledyke 2008). Moreover, food loss (FL) is often used to classify materials intended for human consumption that are subsequently lost, discharged, degraded or contaminated. FAO (Food Agriculture Organisation) of the United Nations defined FL as “any changes in the availability, edibility, wholesomeness or quality of edible material that prevents it from being consumed by people” (Giroto et al. 2015).

The purpose of this study was to develop students’ knowledge and identifying behavioural changes in relation to the waste produced and more specific to the water plastic bottle of 500 ml.

Materials and methods

Area description

The municipality of Paralimni which is located in the eastern region of Cyprus has a permanent population up to 18,601 according to the last inventory studied from the Cyprus Statistical Services on November 2011. The municipality consists the main

economical lung of the island due to the fact that in this area, there are largest hotels' resorts (15% of the total existing in the island); the permanent population increases to 50,000/day from April to October every year (Zorpas et al. 2011), producing more than 15,000 t/year of waste. It has governmental and private schools. Four primary schools, one gymnasium, one lyceum and one technical school belonging to the government and 1 private school as well as several kindergartens.

Schools are working for 180 days/year approximately. According to the lasted waste compositional analysis (Zorpas et al. 2015) which took place for a period of 12 months found that 26% products that can be composted (green waste, yard waste, fruits and vegetables), 20% were food waste, almost 12% toilets and kitchen papers including sanitary, 10.56% papers, 9.41% PMDs, 4.42% plastic films and 5.33% glasses, 2.26% plastics non-recyclable and 0.84% aluminium. Furthermore, in the urban area of the municipality exist 110 recycle points as the municipality participates in the Green Dot (GD) program (plus 240 are in the tourist area). GD Cyprus, raise awareness concerning recycling all around Cyprus and at the same time encouraging volunteerism. Moreover, in yearly base several conferences are organized in order to promote waste prevention. Those conferences focuses on the reused of materials, food waste prevention, home composting, recycling etc. and are given in specific target groups like kids in primary or secondary schools as well as on small medium enterprises, hotels and restaurants owners etc.

From the total compositional analysis, the PMD fraction (Fig. 1) mainly consists of plastics bottles of 500 ml and pots which were equal to 6.05%. The main producer of those plastics bottles are the kids as the parents use to buy every week boxes of water to have in their houses and to be used by the kids when they went to school, or when they went to do sports or other activities after school.

Measuring waste prevention

There is a general environmental education plan that the teachers follow according to the requirements of the Ministry of Education but there is no any specific strategic plan that schools must follow on the framework of waste prevention, although there is a Governmental Strategic Prevention Plan

since 2013 covering the requirements of the waste framework directive 2008/98 (Shar et al. 2008; Zorpas et al. 2014).

In this research, 998 students from the four primary schools were participated. The main problem was the plastic bottle of 500 ml that is produced as a waste every day. Students (ages 6–12) every day have one plastic bottle with water. There are four main categories of students (i) students that belongs on class A and B (ages 6–8) and all of them have either a plastic bottle of water (500 ml not refillable), either a plastic refilling bottle of water; (ii) students that belongs on class C and D (ages 8–10) which most of them have a plastic bottle of 500 ml (not refillable) but few of them have a refilling bottle (as they are shy); (iii) students that belongs to class E and F (ages 10–12) which most of them either bring their plastic bottle of water (500 ml not refillable) or most of them use to buy at least one bottle of water) and (iv) students from all ages that always have their own refilling plastic bottle of water with them.

The evaluation procedure was mainly carry out by using hybrid methods approach as described by Zorpas and Lazardidi (2013). The teachers at the beginning of this research explained to the parents' council about the benefits of the prevention activities as well as about the social and environmental impacts of the daily used of the water plastic bottle of 500 ml. Moreover, it was indicated to parents how expensive is to buy every day and for 180 days at least one bottle of water of 500 ml, as well as about the negative impact of the environment in case that those bottles do not disposed into the recycling bins. Furthermore, it was explained that plastics water bottles generate more than 121 million t of waste each year, 47 milion gallon of oil are needed to create plastic water bottles as well as more than 90% of the trash in the ocean is from plastics.

Nine hundred ninety-eight refilling stainless steel water refilling bottles (SSWRB, of 600 ml—Table 1) shared to each student. For 4 weeks and every day, the teachers measure attitudes and behaviour using a daily questioner with four specific questions: (Q1) which children bring their own plastic bottles, (Q2) how many of them refill the same bottle, (Q3) how many used to buy water from the school and (Q4) how many bring their own water bag (any container). Then for a period of 4 weeks and daily (and after a specific information that were given to the students), the teachers asked the children the same four questions (Q1–Q4), as well as if they bring with them the

Fig. 1 PMD waste stream compositional analysis

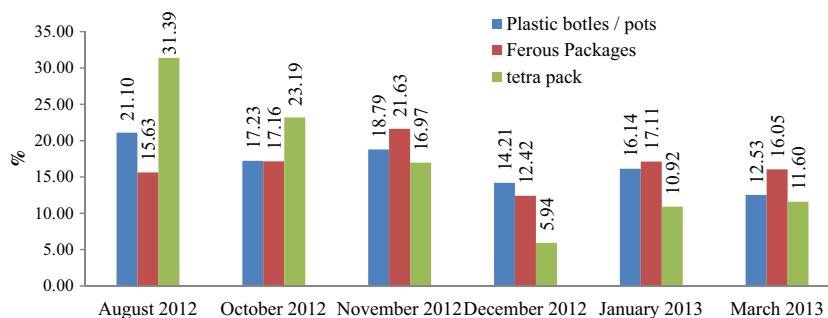


Table 1 Primary schools in Paralimni Municipality

Primary school	No. of students
1st primary school	250
2nd primary school	253
3rd primary school	222
4th primary school	273

Each school has six classes: class A (ages 6–7), class B (ages 7–8), class C (ages 8–9), class D (ages 9–10), class E (ages 10–11) and class F (ages 11–12)

specific refilling stainless steel bottle that were given to them (Q5). Statistical analysis was done using SPSS v2.0.

Results and discussions

With the yearly increasing number of students in the primary schools, the amount of all kinds of wastes is continually increasing every single day. The producer of waste typically originates from the canteens, from kids (related with what they bring with them every day), and administration blocks. Students, teachers, staff and visitors are predominantly the generators, with students be in the first place. More than 40% of the kids from all ages used to bring their own plastic bottles (Q1) which after they discharged as waste, indicate that they are not familiar with recycling. Almost 10% of all ages used to refill their water container from the school. Less than 2% of the students from class A buy water from the school but on the other hand, more than 20% of the class F use to buy every day at least one bottle of water from canteens schools. As indicated in the evaluation period (Figs. 2 and 3), the smaller kids (class A and B are used to bring their own small plastic bottles or their own water plastic bags) while the biggest ages (especially class E and F) used to have their own plastic small bottles (of 500 ml) and do not bring their own water plastic bottle, while at the same time they used to (if they need more water or if they forgot to bring their own water) to buy water from the school. This behaviour is happening for two main reasons: (i) the parents

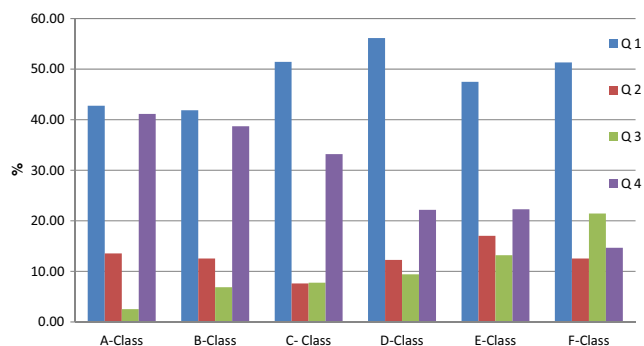


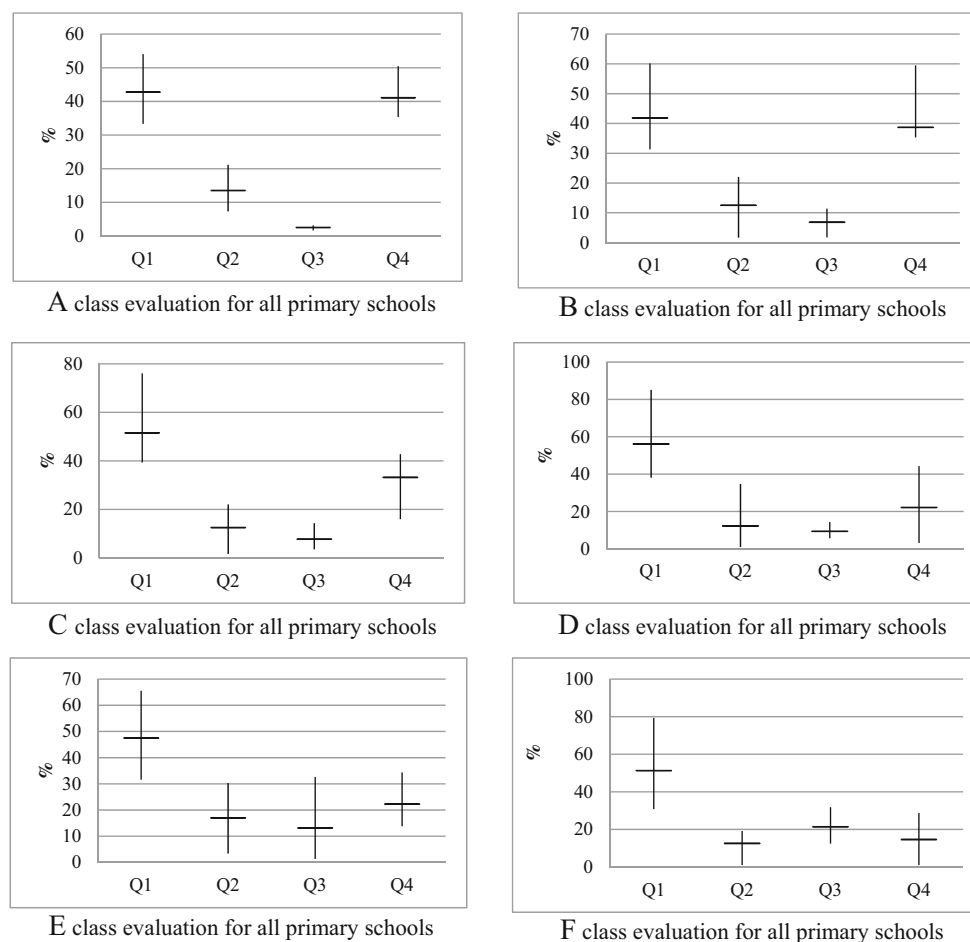
Fig. 2 Comparisons of students' behaviours per class and for all the four primary schools and for the period of 4 weeks

used to give money (approximately €2–3/day) to the kids to have with them if they need to buy something from the school (water, milk, juice, biscuits, cookies, ham and cheese toast, sweetness, candies etc) and (ii) due to the fact that older kids (Class E and F) are feel shame to bring their own water plastic bags and they preferred to buy, indicated to the others that “we are getting teenagers” and “we are controlling ourselves”.

When kids are growing up, their behaviour and attitude changed; from 42% of the participants in class A, bring their own water bottle (any kind of water container–Q4), only 13% of the kids from class F continue to do that. If we look deeper why this continues to happen, we will find out that (a) economic crisis is one of the main reason as to buy water (at least one bottle) cost 0.5€/day/bottle; (b) students are continuously educated by their parents and (c) they start thing and behave more “green”. Through this research, it was found out that a student will cost to each family at least €90 ± 7.5 every school year if they used to buy at least one bottle of water and only hours that the kid staying in the school, producing at the same time approximately 120–200 plastic items of plastic bottle per school year and per kid. Additionally, as most of the kids (90%) have usually external school activities like sports, dancing, foreign languages etc., this quantity is much higher (sometimes could be triple).

The main reason that smaller kids (class A and B from all schools) used to bring their own plastic bottles or their own water plastic container is due to the fact that parents wants to be sure that their kids, they will have during the day their own water as they feel more safe. Essentially, the parents every morning used to prepare the school bag for their kids and which they add sandwich, water, juice and/or milk. They do not feel safe to provide them with money to buy something from the school canteens for many reasons (the most important are that usually they lost their money or they afraid that someone may stole it from them). One more significant reason is due to the fact that, private companies commercialised very smart the “game” of selling “more safe drinkable water” than the drinkable water which exist free in each school in central taps or directly from the water sources given by the municipality. It is quite easy to declare (without specific evidences) that the free drinkable water is affected from pathogen microorganism like total and faecal coliform as well as *Escherichia coli*. This is usually occurring (Shar et al. 2008) because water from different sources i.e. lakes, rivers, reservoirs, groundwaters or from desalination process is subjected to varying degrees of faecal pollution, and therefore freshwater is a path of transmission of many pathogenic bacteria, protozoa and virus. The safe supply of water is the responsibility of the water department Authorities and the municipality and this based on Human Consumption Water Law; the water must be free of pathogens. Specifically, the concentration of total coliforms must be less than 10 cfu/ml while the other microorganism like *Staphylococcus aureus*, *E. coli*, faecal streptococci and *Pseudomonas aeruginosa* must not be identified (National Law 2001).

Fig. 3 Evaluation of students' behaviours per class and for all the four primary schools and for the period of 4 weeks



Figures 4 and 5 present the results from the second evaluation period and after the teachers and the kids were totally informed regarding the benefits of using the same water plastic container. It was carefully explained to parents and teachers that plastic bottles are usually presented with bisphenol A (BPA) (Cooper et al. 2011). Bottled water has become very common as it is perceived as source of clean water. Nevertheless, due to its low diffusivity and chemical inertness (Franz and Welle 2002), PET (polyethylene terephthalate) is the most popular material for food or beverage packaging, accounting for >99% of all beverage bottles (ILSI 2000). However, recent studies demonstrated that PET bottles can release harmful chemicals such as antimony (Sb) and BPA into water (Westerhoff et al. 2008; Santhi et al. 2012). BPA in PET bottled

water has been mentioned by Li et al. (2010). BPA is mainly used extensively in the production of polycarbonate plastics, consumer goods, epoxy resins and coatings used to line metallic food and beverage cans (Cooper et al. 2011; Fan et al. 2014). There is great worry regarding the possible harmful effects of BPA discharge into foods and beverages from packaging containers (mainly plastics). At room temperature, the concentration of BPA migrating from polycarbonate bottles ranged from 0.2 to 0.3 mg/l. Examination of BPA in urine samples indicated that BPA is present in over 93% of the US population (Calafat et al. 2005). The mean urine BPA concentrations in various US populations are found to be in the range of 1–6 µg/L, or nearly 6–26 nM (Calafate et al. 2005; Cooper et al. 2011). BPA has been exposed to impact the actions of endogenous estrogenic

Fig. 4 Comparisons of students' behaviours per class and for all the four primary schools and for the second period of 4 weeks (with the SSWRB)

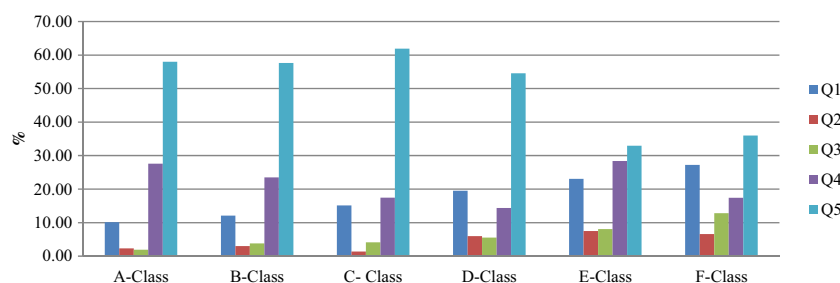
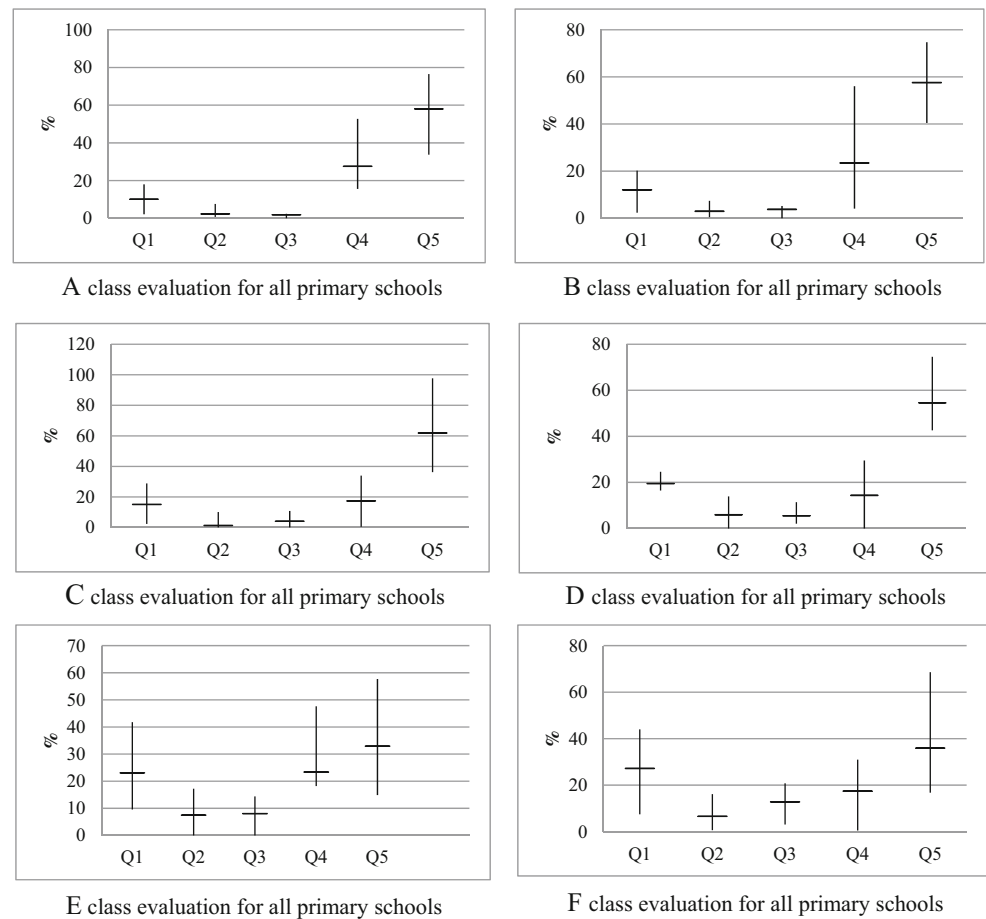


Fig. 5 Evaluation of students' behaviours per class and for all the four primary schools for the second period of 4 weeks (with the SSWRB)



steroid hormones (e.g. 17 β -estradiol, estrone or estriol), but the full extent of its effects are not yet been clearly understandable (Cooper et al. 2011).

From the evaluation, it can be concluded that after the specific information is given to the teachers, parents and students from all ages accept the SSWRB but mostly the smaller kids (A class 59% and then B class 58%) are more receptive and enthusiastic for something new. Impressing was that the biggest kids (E and F class), they accept to use this new SSWRB in more than 30%, and they did not buy water from the school (the percentage of buying water from the school from the first to second week was reduce from 13 to 8% regarding class E and from 21 to 2% for class F). However, 90% of the kids in class E and 87% from class F, as they behave (or want to be presented) as adolescents, they chose to reuse their own plastic bottle, or to bring their own water plastic container or to have the new SSWRB. During the first evaluation period of the evaluation, a total amount of 48% of the kids from all ages discharged their plastic bottles to the waste bins. This amount was reduced to 18% for all the kids. At the same time, more than 60% of the kids from all ages accept to prevent using either their own water containers or the SSWRB.

A very important result is that, all students during the second evaluation period (Fig. 5, Q2) did not fill up the same plastic

bottle that they used to bring with them. From the first period, the total percentage from all ages and from all the schools (Fig. 3) used to refill the same plastic bottle was $12.66 \pm 5.24\%$; while during the second evaluation period (Fig. 5), this was reduced to $4.48 \pm 2.74\%$. This is due to the fact that during the second evaluation period, it was totally explained that usually if they used those plastic bottles without washing them, those will present several problems like high amount of bacteria like heterotrophic bacteria and total coliforms which usually are more than 10 cfu/100 ml (Oliphant et al. 2002). Those microorganisms (Oliphant et al. 2002) are the results of the absence of personal hygiene; like insufficient and inadequate hand washing after students have used the bathroom services could result in faecal coliforms in the schoolrooms.

The main aim of the environmental education is not only to increase conscious eco-citizens who comprehend the nature as a whole, develop a critical approach in their interaction with environment and are sensitive about environmental issues, but also to help individuals figure out and make researches about environmental issues as well as in their efforts to solve the problems and take decisions as to the matter (Godwell and Ekpe 2011). The importance of environmental education program in schools, to prevent and solve environmental issues and problems as well as to increase sensitive and conscious

citizens about environment becomes essential and important. The schools plays significant role in providing this education. The attitudes of teachers at schools affect the responses of the students, too. Thus, it is vital for teachers to be an exemplary model (Littledyke 2008). Schools according to Bell (2002) are no longer expected simply to produce general development plans that focus on any aspect of the school's work that might be identified as a priority at that time. School goals must derive from national ones for similar schools. The plans into which these goals are combined must focus on strategies for bringing about curriculum change that will lead to improvements in public performance. Environmental awareness must developed and apply in any school as humanity suffers from many environmental problems (air pollution, waste production, water pollution, climate changes etc.) which need to be tackled at the individual level, requiring individuals to develop those attitudes which will guide them to environmentally supportive behaviour (bin Hamad Al-Rabaani and Al-Mekhlafi 2009). Formation and change of attitude are complicate. People (of all ages) are always modifying, adopting and abandoning attitudes to fit the ever changing needs, benefits and interests. Attitude cannot be improved and changed by simple training or education. Acceptance of new attitude mainly depends on who is presenting the knowledge, how the person is perceived, how it is presented, the credibility of the communicator, and the conditions by which the knowledge was received. Researches in social sciences has exposed and indicate that, knowledge on a topic may increase; people could even change attitudes, but the step to improved personal behaviours and practises is depending on psychological factors which is a complex social set. Opinions, behaviours, and attitudes that are satisfied and reinforced are likely to be repeated and combined into our personal value set (Curzon 2003). Quite often our attitudes about politics as well as about environment come from information and persuasive communications (Johnston 2010) and been affected from our social environment and our families believes (sometimes).

Conclusions

Waste prevention encompasses a range of policy options and has a broad range of benefits. Targeting at-source waste creation, it reduces the amount and toxicity of waste before composting, recycling, energy efficiency and landfilling become options. Waste prevention includes measures to reduce the adverse impacts of the produced waste on the environment and human health. The waste minimization prevention campaign is in priority in order to achieve all the proposed activities and especially in young ages. Public awareness event must be taken into consideration before the establishment of any prevention action. It is not easy to change people's behaviour without any specific awareness event. To move the

agenda forward on prevention, particular emphasis on behavioural change must be done starting from young ages. Moreover, behaviours are complex and non-linear. Each behaviour is determined by various (often interrelated) factors, many of which need addressing simultaneously to facilitate change. Furthermore, different audiences, different segments of society in relation to environmental issues behave differently, and require targeted and/or tailored interventions.

Acknowledgements This paper presents the results from a large LIFE + Project which focuses on the Development and Demonstration of a Waste Prevention Support Tool for Local Authorities WASP Tool (Project LIFE10 ENV/GR/000622, that is cofunded from EU).

References

- Bell L (2002) Strategy planning and school management: full of sound and fury, signifying nothing. *J Educ Admi* 40(5):407–424
- bin Hamad Al-Rabaani A, Al-Mekhlafi SSM (2009) Attitudes of Sultan Qaboos university students towards some environmental problems and their willingness to take action to reduce them. *J Social Sci* 5(1): 9–15
- Calafat AM, Kuklenyik Z, Reidy JA, Caudill SP, Ekong J, Needham LL (2005) Urinary concentrations of bisphenol A and 4-nonylphenol in a human reference population. *Environ Health Perspect* 113:391–395
- Cooper EJ, Kendig LE, Belcher MS (2011) Assessment of bisphenol A released from reusable plastic, aluminium and stainless steel water bottles. *Chemosphere* 85:943–947
- Cox J, Giorgi S, Sharp V, Strange K, Wilson DC, Blakey N (2010) Household waste prevention—a review of evidence. *Waste Manage Res* 28:193–219
- Curzon LB (2003) Teaching in further education: an outline of principles and practice. Cornwall: MPG Books Ltd.
- Daddi T, Frey M, Rosa De Giacomo M, Testa F Fabio Iraldo (2015) Macro-economic and development indexes and ISO14001 certificates: a cross national analysis *J Clean Prod* 108:1239–1248.
- de Kumar A (2004) Environmental education. New Age International, New Delhi
- European Parliament and the Council of the European Union (2008) Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain directives. *Official Journal* 312:3–30
- Fan YY, Zheng LJ, Ren HJ, Luo J, Cui YX, Ma QL (2014) Effects of storage temperature and duration on release of antimony and bisphenol A from polyethylene terephthalate drinking water bottles of China. *Environ Poll* 192:113–120
- Fell D, Cox J, Wilson DC (2010) Future waste growth, modelling and decoupling. *Waste Manage Res* 28:281
- Foo YK (2013) A vision on the role of environmental higher education contributing to the sustainable development in Malaysia. *J Clean Prod* 61:6–12
- Franz R, Welle F (2002) Recycled poly(ethylene terephthalate) for direct food contact applications: challenge test of an inline recycling process. *Food Addit Contam* 19:502–511
- Giroto F, Alibardi L, Cossu R (2015) Food waste generation and industrial uses: a review. *Waste Manag* 45:32–41
- Godwell N, Ekpe I (2011) Framework and tools for environmental management in Africa. Dakar: CODESRIA (2011).
- Haigh M (2009) Greening the university curriculum: appraising an international movement. In: Chalkey B, Haigh M, Higgitt D (eds) *Education for sustainable development, Papers in honour of the*

- United Nations decade of education for sustainable development. Routledge, London, pp 25–42
- Hume T, Barry J (2015) Environmental education and education for sustainable development. *Int Enc Soc Behav Sci* 2:733–739
- ILSI (2000) Packaging materials: polyethylene terephthalate (PET) for food packaging applications. International Life Sciences Institute (ILSI), Europe, Belgium
- Jickling B, Wals AEJ (2008) Globalization and environmental education: looking beyond sustainable development. *J Curriculum Stud* 40(1): 1–21
- Johnston H (2010) Shaping beliefs and attitudes: a handbook of attitude change strategies http://www.coedu.usf.edu/main/departments/seced/Faculty/documents/AttitudeChapter_H_johnston.pdf (accessed on December 2016)
- Kanyimba TA, Richter WB, Raath PS (2014) The effectiveness of an environmental management system in selected South African primary schools. *J Clean Prod* 66:479–488
- Kopnina H (2012a) Education for sustainable development (ESD): the turn away from environment in environment education? *Environ. Education Res* 18(5):699–717
- Kopnina H (2012b) Evaluating education for sustainable development (ESD): using ecocentric and anthropocentric attitudes towards sustainable development (EAATSD) scale. *Environ Dev Sustain* 14(4). doi:10.1007/s10668-012-9395-z
- Kopnina H (2014) Education for sustainable development (ESD) as if environment really mattered. *Environ Dev* 12:37–46
- Lasaridi K, Chroni C, Abeliotis K, Kyriacou A, Zorpas AA (2014) Environmental assessment of composting in the context of sustainable waste management. In: Zorpas AA (ed) Sustainability behind sustainability. Nova Science Publishers Inc, New York, pp 229–242
- Li X, Ying GG, Su HC, Yang XB, Wang L (2010) Simultaneous determination and assessment of 4-nonylphenol, bisphenol A and triclosan in tap water, bottled water and baby bottles. *Environ Int* 36:557–562
- Littledyke M (2008) Science education for environmental awareness: approaches to integrating cognitive and affective domains. *Environ Educ Res* 14(1):1–17
- Nag A, Vizayakumar K (2005) Environmental education and solid waste management. New Delhi: New Age International.
- National Law (2001) Quality of Water for Human Consumption (Monitoring and Control) N87(I)/2001, [http://www.moa.gov.cy/moa/wdd/wdd.nsf/All/FD9B61489563B5FFC2257403003FC3E4/\\$file/N.87_I_2001.pdf?OpenElement](http://www.moa.gov.cy/moa/wdd/wdd.nsf/All/FD9B61489563B5FFC2257403003FC3E4/$file/N.87_I_2001.pdf?OpenElement) (accessed on January, 2015); 537–558.
- Oliphant JA, Ryan MC, Chu A (2002) Bacterial water quality in personal water bottles of elementary students. *C J Public Health* 93(5):366–367
- Read M, Gregory KM, Phillips SP (2009) An evaluation of four key methods for monitoring household waste prevention campaigns in the UK. *Resour Conserv Recy* 54:9–20
- Rickinson M, Lundholm C, Hopwood N (2009) Environmental Learning. Insights from Research into Student Experience. Springer Dordrecht Heidelberg, London
- Santhi VA, Sakai N, Ahmad ED, Mustafa AM (2012) Occurrence of bisphenol A in surface water, drinking water and plasma from Malaysia with exposure assessment from consumption of drinking water. *Sci Total Environ* 427–428:332–338
- Selby D (2006) The firm and shaky ground of education for sustainable development. *J Geogr Higher Educ* 30(2):351–365
- Shar HA, Kazi FY, Zardari M, Soomro HI (2008) Enumeration of total and fecal coliform bacteria in drinking water of Khairpur Sindh. *Pak J Med Res* 47(1) <http://www.pmr.org.pk/fecalcoliform.htm>
- Sharp V, Giorgi S, Wilson DC (2010) Methods to monitor and evaluate household waste prevention. *Waste Manage Res* 28:269–280
- Testa F, Daddi T, Rosa De Giacomo M, Iraldo F, Frey M (2014a) The effect of integrated pollution prevention and control regulation on facility performance. *J Cleaner Pro*
- Testa F, Rizzi F, Daddi T, Marzia Gusmerotti N, Frey M, Iraldo F (2014b) EMAS and ISO 14001: the differences in effectively improving environmental performance. *J Clean Prod* 68(1):165–173
- UNESCO (2006) Framework for the UNDES International Implementation Scheme. UNESCO. Available from: <http://unesdoc.unesco.org/images/0014/001486/148650E.pdf>.
- Voukkali I, Loizia P, Zorpas AA (2014) Definitions of sustainability. Inc Sustainability Behind Sustainability, In: A.A. Zorpas (Ed), Nova Science Publisher, New York, pp 1–7.
- Westerhoff P, Prapaipong P, Shock E, Hillaireau A (2008) Antimony leaching from polyethylene terephthalate (PET) plastic used for bottled drinking water. *Water Res* 42:551–556
- Zorpas AA, Lasaridi K (2013) Measuring waste prevention. *Waste Manag* 33:1047–1056
- Zorpas AA, Coumi C, Drtil M, Voukkali I (2011) Municipal sewage sludge characteristics and waste water treatment plant effectiveness under warm climate conditions. *Desalin Water Treat* 36:1–15
- Zorpas AA, Lasaridi K, Abeliotis K, Voukkali I, Loizia P, Fitiri L, Chroni C, Bikaki N (2014) Waste prevention campaign regarding the waste framework directive. *Fresen Environ Bul* 23(11a): 2876–2883
- Zorpas AA, Lasaridi K, Voukkali I, Loizia P, Chroni C (2015) Household waste compositional analysis variation from insular communities in the framework of waste prevention strategy plans. *Waste Manag* 38: 3–11